Kalvakuntla Ritanshu Kumar

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Education

Vellore Institute of Technology (VIT) Vellore, India
Integrated M.Tech in Computer Science and Engineering (CGPA: 8.06) Aug 2022 – Sept 2027
Specialization: Data Science

Experience

Machine Learning Intern June 2025 – July 2025
MANMUN PVT LTD, Hyderabad

- Worked on AI-driven application development focusing on Machine Learning and Deep Learning, with practical exposure to implementing and testing ML models.
- Built a Document Q&A RAG Assistant using Python, ChromaDB, Llama 3.1/Groq API, SQLite and Streamlit, with retrieval evaluation and response-quality testing.

Projects

Document Q&A RAG Assistant | *Python, ChromaDB, Llama 3.1/Groq API, SQLite, Streamlit* |

- Built a retrieval-augmented document QA system using ChromaDB vector search and Llama 3.1 via Groq API for grounded responses.
- Implemented document ingestion, chunking, embedding retrieval, prompt construction, and SQLite-based evaluation logging.

Deep Learning Framework for Rice Leaf Disease Classification | *Python, PyTorch, MobileNetV3, Self-Attention* |

- Developed a MobileNetV3 Large classifier with self-attention, transfer learning, and multi-scale features across 10 rice disease classes.
- Achieved 99.80% validation accuracy and 99.81% F1-score; achieved 96.35% accuracy on the out-of-domain Paddy Doctor dataset.

Hybrid CNN-Transformer Model for Pneumonia Detection | *Python, PyTorch, Flask, Grad-CAM* |

- Built HybridXRayNet using multi-scale CNN blocks, Transformer Encoder, adaptive fusion, and Grad-CAM for interpretable diagnosis.
- Achieved 92.6% Recall, 87.2% F1-score, and 0.918 ROC-AUC on 624 test images, improving sensitivity over ResNet-50.

CAPTCHA Recognition using CRNN + CTC | *Python, PyTorch, ResNet18, CRNN, CTC, Computer Vision* |

- Developed end-to-end CAPTCHA recognition using ResNet18 feature extraction, recurrent sequence modeling, and CTC decoding without segmentation.
- Implemented vocabulary generation, train/test splitting, sequence labeling, and CTC-based training and decoding in PyTorch.

Technical Skills

Programming & Data Science: Python, SQL, C++, JavaScript, Pandas, NumPy, Scikit-learn, EDA, Statistics, Feature Selection, Data Visualization, Power BI

AI/ML & Deep Learning: PyTorch, TensorFlow/Keras, CNNs, Transformers, OpenCV, NLP, Computer Vision, LLMs, RAG, Embeddings, Vector Databases, Prompt Engineering

Web Development & APIs: React.js, FastAPI, Flask, Streamlit, Node.js, Express.js, MongoDB, PostgreSQL

Tools & DevOps: Git, GitHub, Docker, Apache Airflow, AWS (EC2, S3), Vercel

Certifications

- **100x Dev**: Live Full Stack Open Source Cohort-1
- **Coursera**: Supervised Machine Learning
- **Forage**: Tata GenAI Powered Data Analytics Job Simulation
- **Embrizon**: Machine Learning with Python